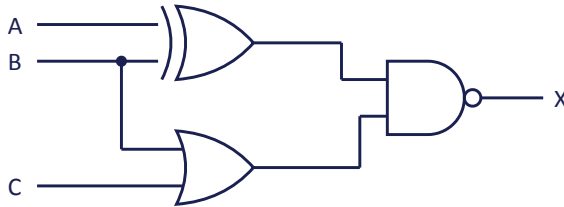


Logic Circuits — Practice · AS 9618 — ANSWERS (student copy) · first edition, the sheet set as homework

1. $H = ((\text{NOT } G) \text{ AND } L) \text{ OR } T$ — 1 mark NOT applied to G only, 1 mark structure (AND inside, OR outside). Accept equivalent bracketing.
2. 1 mark per gate-cluster: $\text{XOR}(A,B) \cdot \text{OR}(B,C)$ with a **junction dot** where B branches · NAND joining them (bubble on the output).



Reject: B written twice as two separate inputs, branch without a dot, or the NAND bubble missing/floating.

3. $X = (A \text{ XOR } B) \text{ NOR } (\text{NOT } C)$ — 1 mark both branches named (XOR arc + NOT), 1 mark NOR as the final/middle operator. Outside-in check: last gate NOR $\rightarrow$ sentence shape (...) NOR (...).
4. Working columns A NAND B: 1, 1, 1, 1, 1, 1, 0, 0 · A OR C: 0, 1, 0, 1, 1, 1, 1, 1; X rows 000 $\rightarrow$ 111: 0, 1, 0, 1, 1, 1, 0, 0 — 1 mark per 2 correct X rows.
5. Two 1s, but in the *equal-inputs* seats (0 0 and 1 1) — no single gate fits; recipe time: $X = ((\text{NOT } A) \text{ AND } (\text{NOT } B)) \text{ OR } (A \text{ AND } B)$. Also accept NOT (A XOR B) / XNOR. 2 marks any correct form; 1 mark for a correct method with one slip.
6. (a) By the **tail** (back edge): **straight $\rightarrow$ AND family, curved $\rightarrow$ OR family**; XOR needs its second arc visibly separate [1]. (b) The wires are **not connected** — a crossing without a junction dot is two independent wires [1].

Total: 15 marks. Q2 is the dot question — tail first, dot always. Q5 pays the count-the-1s lesson: the two-1s pattern only means XOR when the 1s sit in the different-inputs rows.

🕒 **The quiet one, unhinted.** (A XOR B) opens **both** question 2 and question 3 — drawn in one, read in the other — and question 5's answer works out to **XNOR**, which is **exactly its negation**. XOR is in three of the five questions, twice as itself and once upside down, and the sheet never mentions it once.